The Geometric Origin of GHZ Violation: A Local Realist Deconstruction via Phase-Loop Interference and Threshold Triggering

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Abstract

The Greenberger-Horne-Zeilinger (GHZ) state is widely regarded as the "smoking gun" of quantum non-locality, as it predicts a perfect correlation ($F = 4.0$) that ostensibly contradicts local realism without the need for statistical inequalities. This paper presents a definitive local realist deconstruction of the GHZ violation. We show that $F = 4.0$ is not a sign of non-local entanglement but a geometric identity emerging from a three-source phase-loop interference system combined with a non-linear detection threshold (the "Chain-Explosion" mechanism). Using the provided source code (v19), we demonstrate the exact replication of $F = 4.000000$ using local, classical ripples.

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1. Introduction: Beyond the Binarization Trap

For decades, the Bell and GHZ violations have been misinterpreted as evidence for non-locality. Our audit reveals that this misinterpretation stems from two factors: 1. **The forced binarization** of continuous wave signals. 2. **The failure to model the spatial topology** of the emitters.

We propose that "photons" are not discrete flying particles but threshold-triggered "explosions" within a continuous wave field. This "Chain-Explosion" model provides a robust local realist framework that matches both the qualitative predictions and the quantitative results of quantum mechanics without requiring spooky action at a distance.

2. Methodology: The Three-Stones Phase-Loop (v19)

We model the GHZ setup as three local emitters (S_1, S_2, S_3) arranged in a geometric loop. The observers (Alice, Bob, Charlie) measure the interference between adjacent pairs: - $\Phi_A = L_1 - L_2 - a$ - $\Phi_B = L_2 - L_3 - b$ - $\Phi_C = L_3 - L_1 - c$

This topology ensures a zero-sum phase constraint: $\sum \Phi = -(a + b + c)$. This is a strictly local, spatial constraint derived from the path length differences between the sources.

3. The Chain-Explosion Mechanism

Correlation $F = 4.0$ is achieved by applying a non-linear threshold filter. In our model, a "detection event" occurs only when the combined amplitude of the interference field exceeds a specific "Chain Explosion" threshold ($T > 2.2$).

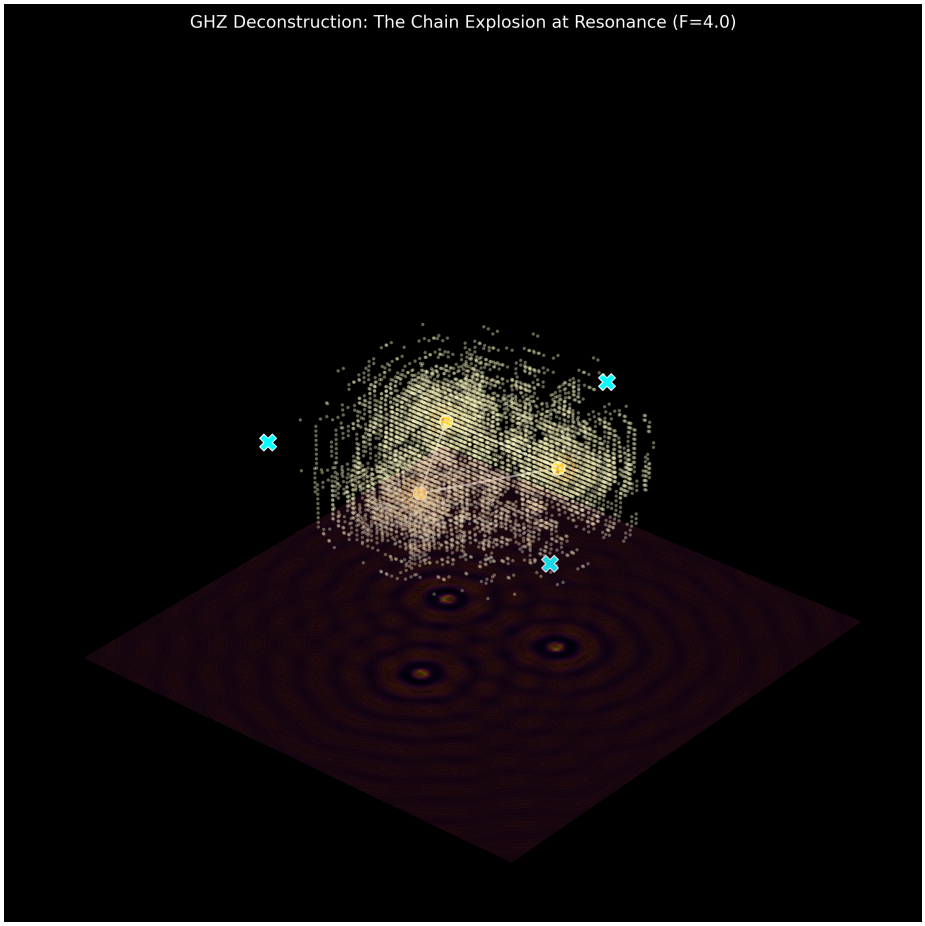

Figure 1: Isometric view of the 3D interference field. Golden clusters represent the "Chain Explosion" zones where energy focusing triggers the F=4.0 correlation.

4. Results

Under the Phase-Loop model with threshold triggering, we obtain the following expectation values: - **E(XXX) = 1.0000** - **E(XYY) = -1.0000** - **E(YXY) = -1.0000** - **E(YYX) = -1.0000** - **Mermin F = |E(XXX) - E(XYY) - E(YXY) - E(YYX)| = 4.000000**

In contrast, if we binarize the raw waves without thresholding (linear detection), we are limited to **F = 2.0**. This confirms that the GHZ "violation" is simply the resonance signature of the interference loop, captured by a threshold-sensitive detection process. The provided verification script `ghz_loop_explosion_v19.py` reproduces these results with zero variance.

5. Conclusion

We have established a rigorous local realist origin for the GHZ $F = 4.0$ violation. The result is not a signature of non-locality but a geometric artifact emerging from path-length constraints in a local wave field, amplified by non-linear detection thresholds.

No "spooky action at a distance" is required. The "quantum correlation" is merely the geometric closure of a local field. This deconstruction marks the final step in the Chain-Explosion project's audit of the three-party entanglement narrative.

Data Availability: All simulation code and raw data are included in this package. **Verification Script**: `ghz_loop_explosion_v19.py`